

Persistent Cookie Analyzer with a Graphical User Interface

Dr. Polasi Sushma, Chegondi Akhil Sampath Vinay, Gongalla Trisha Reddy, Sanika Praveen

Abstract :

Persistent cookies, often overlooked in web interactions, pose significant privacy and security risks due to their prolonged storage and potential misuse. This research introduces the Persistent Cookie Analyzer with a Graphical User Interface, a comprehensive software tool designed to analyze and assess the risks of persistent cookies in web browsers. The tool focuses on detecting cookies from local chrome databases, evaluating their attributes for vulnerabilities, and ensuring compliance with privacy regulations like GDPR (General Data Protection Rule) and CCPA (California Consumer Privacy Act). It incorporates an intuitive graphical user interface (GUI) built with Python's Tkinter, allowing users to interact seamlessly and analyze cookies without requiring technical expertise. Key functionalities include extracting persistent cookies from chrome storage, identifying vulnerabilities such as missing Secure and HttpOnly flags, and evaluating compliance with cybersecurity standards like OWASP and NIST. The tool assigns a risk score to each cookie based on critical attributes such as secure transmission, expiration date, SameSite policy, and cross-site accessibility, providing users with actionable insights. It generates detailed PDF reports summarizing risks, vulnerabilities, and compliance findings, catering to both individual users and organizations striving for enhanced privacy and security practices. This paper outlines the architecture, implementation of the Persistent Cookie Analyzer, emphasizing its role in mitigating cookie-related risks. By combining technical rigor with user-friendly design, the tool empowers users to identify and manage persistent cookies effectively, contributing to a safer digital ecosystem. Future enhancements aim to include support for additional browsers, real-time cookie monitoring, and artificial intelligence and machine learning based risk prediction for advanced threat assessment.

Keywords : SameSite flag, HttpOnly, Secure flag, Risk Assessment, GDPR, CCPA.

1. Introduction

Cookies are small blocks of data generated by web servers to enhance the browsing experience by storing user-specific information such as authentication tokens, preferences, and session data. They play a crucial role in modern web applications, improving user convenience and functionality. Among the different types of cookies, session cookies and persistent cookies are the most well-known. Session cookies exist temporarily while a user navigates a website and are deleted once the browser is closed. In contrast, persistent cookies remain stored on the user's device for a predefined lifespan set by the website, enabling data retention across multiple sessions. Other types of cookies, such as HttpOnly cookies, Secure cookies, SameSite cookies, third-party cookies, and zombie cookies, serve various functions and come with their own security and privacy considerations. Persistent cookies, while beneficial for user experience, pose significant security and privacy risks. They allow websites and third-party entities to track users across multiple browsing sessions, raising concerns about unauthorized data collection and

user profiling. Additionally, improperly configured persistent cookies can be vulnerable to attacks such as cross-site scripting (XSS), cross-site request forgery (CSRF), and session hijacking. Weak implementation practices, including the absence of Secure or HttpOnly flags, improper SameSite attributes, and overly permissive domain settings, further increase their risks. Moreover, persistent cookies may lead to regulatory non-compliance, especially with privacy laws such as the GDPR and CCPA.

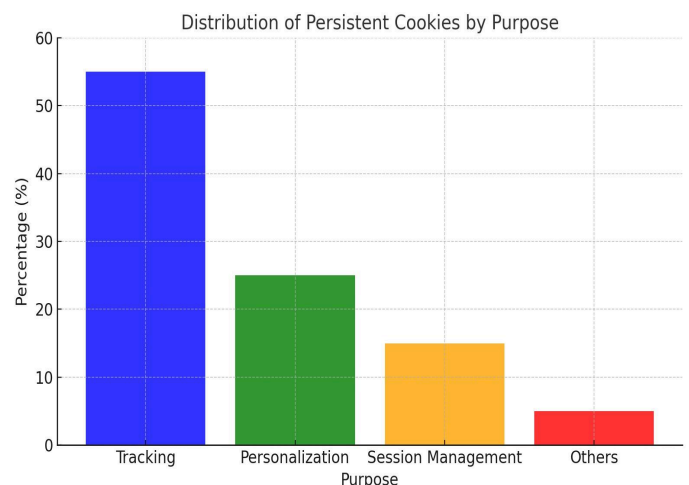


Fig 1. Usage of Persistent Cookie

Although several tools and browser extensions exist for cookie management, they primarily focus on basic functionalities like viewing and deleting cookies. Most of these tools lack comprehensive analysis features and are often designed for developers or cybersecurity professionals, making them inaccessible to non-technical users. The Persistent Cookie Analyzer with a Graphical User Interface addresses these challenges by offering advanced functionalities such as vulnerability detection, risk scoring, detailed analysis of persistent cookies, and compliance assessments. With an intuitive user interface, this tool enables both individuals and organizations to effectively manage cookie-related risks, bridging the gap between technical complexity and practical usability.

1.2 Problem Statement

The widespread use of persistent cookies in web browsing has a significant impact on user privacy, both directly and indirectly. These cookies are often stored on users' devices without their consent, enabling websites to collect data without explicit permission. Additionally, some websites monitor user behavior during browsing sessions, while others fail to properly configure their cookies, leaving them vulnerable to exploitation by attackers. Users, however, often lack the necessary tools and knowledge to manage and assess the security and privacy risks associated with persistent cookies. Existing solutions are typically functionally limited and fail to provide thorough analysis or intuitive user interfaces, making it difficult for non-technical users to understand and address these risks. As a result, there is a critical need for a robust solution that goes beyond simple detection. The ideal tool should offer in-depth analysis of persistent cookies, evaluate the associated risks, implement a risk scoring system, and provide management features. Furthermore, the tool should generate comprehensive reports to empower users to better protect their web privacy.

1.3 Objectives

The Persistent Cookie Analyzer with a Graphical User Interface addresses the need for effective management and analysis of persistent cookies in

web browsers. Its primary objective is to extract and analyze cookies to identify security vulnerabilities and evaluate compliance with data protection regulations such as GDPR and CCPA. The tool assesses critical cookie attributes, including the Secure and HttpOnly flags, SameSite policies, expiration timestamps, and domain scope, to assign a risk score. By detecting cookies with extended lifespans and identifying potential vulnerabilities, it enables users to understand the associated security risks. Designed with an intuitive graphical interface, the tool ensures accessibility for non-technical users while simplifying cookie analysis. Additionally, it generates comprehensive PDF reports that summarize security risks, vulnerability assessments, and compliance findings, empowering users and organizations to make data-driven decisions for enhancing cybersecurity posture and ensuring regulatory adherence.

2. Literature Review

Persistent Cookies play a pivotal role in modern web applications, enabling functionalities such as session management, user authentication, and personalized user experiences. These cookies are stored on the user's device and transmit data to the web server whenever the user visits the website. Persistent cookies remain active until their expiration date, at which point they stop interacting with the server. They are stored in the browser's database under their respective profiles, and users cannot directly analyze them as they are typically encrypted. Despite their utility, the use of persistent cookies raises significant privacy and security concerns. GDPR and CCPA emphasize the importance of cookie transparency and obtaining user consent. Research by Tikkinen-Piri et al. (2018) highlights the impact of GDPR on cookie management, underscoring the need for tools that assist organizations in ensuring compliance. Similarly, research by Degeling et al. (2019) reveals widespread violations of privacy regulations in cookie implementations, emphasizing the necessity for automated tools to assess compliance with privacy standards.

Existing cookie analysis tools, such as browser developer consoles and extensions like Cookie Inspector and EditThisCookie, offer basic cookie

management capabilities. However, they lack advanced features such as vulnerability detection, risk scoring, and regulatory compliance assessments. These tools, while useful for developers, often lack user-friendly interfaces, limiting accessibility for non-technical users and preventing widespread adoption.

Cybersecurity frameworks, such as OWASP and NIST, identify common vulnerabilities in cookie management. These include missing Secure and HttpOnly flags, and improper SameSite attributes, which contribute to risks like cross-site scripting (XSS) and cross-site request forgery (CSRF). Further research by Calzavara et al. (2020) highlights the security risks associated with persistent cookies that have extended expiration periods, as they increase exposure to unauthorized access and exploitation.

Despite the advancements in cookie management and privacy research, existing tools and solutions often fail to provide comprehensive solutions for analyzing persistent cookies specifically. The Persistent Cookie Analyzer with a Graphical User Interface fills this gap by offering a tool that integrates advanced risk analysis, privacy compliance checks, and a user-friendly graphical interface. This tool caters to both technical and non-technical users, addressing critical issues related to cookie analysis, security vulnerabilities, and regulatory compliance, ultimately contributing to enhanced privacy protection and cybersecurity practices.

3. Gap in the Market

As the digital ecosystem grows, cookies have become essential for providing seamless user experiences and personalized services. However, the increasing reliance on persistent cookies, along with rising privacy and security concerns, has created a significant gap in the market for effective cookie management solutions.

While several tools and browser extensions are available for general cookie analysis, they often lack the specialized capabilities needed to assess the risks associated with persistent cookies. Many existing tools focus on basic functions like cookie deletion or flagging, but fail to offer advanced features such as vulnerability detection or compliance evaluation with privacy regulations like GDPR and CCPA. Furthermore, these tools are often designed for technical users,

making them inaccessible to non-technical users and smaller organizations. They also lack detailed reporting and actionable insights necessary for proactive risk mitigation.

Organizations and individuals face challenges in identifying high-risk cookies, understanding their implications, and ensuring compliance with global privacy standards. This gap is further exacerbated by the lack of solutions that integrate persistent cookie detection, risk analysis, and privacy compliance into a single platform with an intuitive graphical user interface (GUI).

The Persistent Cookie Analyzer with a Graphical User Interface addresses this gap by combining advanced cookie analysis with an easy-to-use interface. This tool offers a comprehensive solution for managing and mitigating risks associated with persistent cookies while ensuring compliance with privacy regulations, making it accessible to both technical and non-technical users.

4. Methodology

The Persistent Cookie Analyzer with a Graphical User Interface (GUI) is designed to extract, analyze, filter, and delete cookies from Google Chrome profiles while assessing their security risks. The methodology for implementing the Persistent Cookie Analyzer with a GUI is structured into a systematic workflow designed to ensure the accurate extraction, analysis, and reporting of browser cookies. The process begins with cookie extraction, where cookies are retrieved from the local storage of modern web browsers, such as Google Chrome and Mozilla Firefox. This involves accessing browser-specific SQLite databases and utilizing Python libraries like sqlite3 to query and extract cookie attributes, including name, domain, path, expiration date, and security flags (Secure, HttpOnly, and SameSite).

Following extraction, a risk assessment framework is applied to evaluate cookies based on predefined security criteria. The framework incorporates a scoring system that assigns numerical values to cookies based on their vulnerabilities. For instance, missing Secure or HttpOnly flags and improper SameSite attributes contribute to the overall risk score. This allows users to prioritize addressing high-risk cookies,

enhancing the security posture of their web applications.

To address regulatory requirements, a compliance checker module was conceptualized. This module cross-references cookie attributes against privacy standards such as GDPR and CCPA, as well as cybersecurity frameworks like OWASP and NIST. By implementing rule-based logic, the module ensures that compliance checks are accurate and dynamic, flagging non-compliant cookies and providing actionable recommendations for remediation.

The final stage involves generating actionable insights and reports for users through a Graphical User Interface (GUI). The GUI, developed using Python’s Tkinter library, provides an interactive and user-friendly interface that displays extracted cookies, risk scores, and compliance statuses. Additionally, it allows users to take specific actions, such as deleting cookies or modifying their attributes. The insights are further consolidated into detailed PDF reports using the FPDF library, making the tool accessible to individuals and organizations alike. The methodology emphasizes modularity and scalability, enabling future enhancements, such as real-time monitoring and extended browser support.

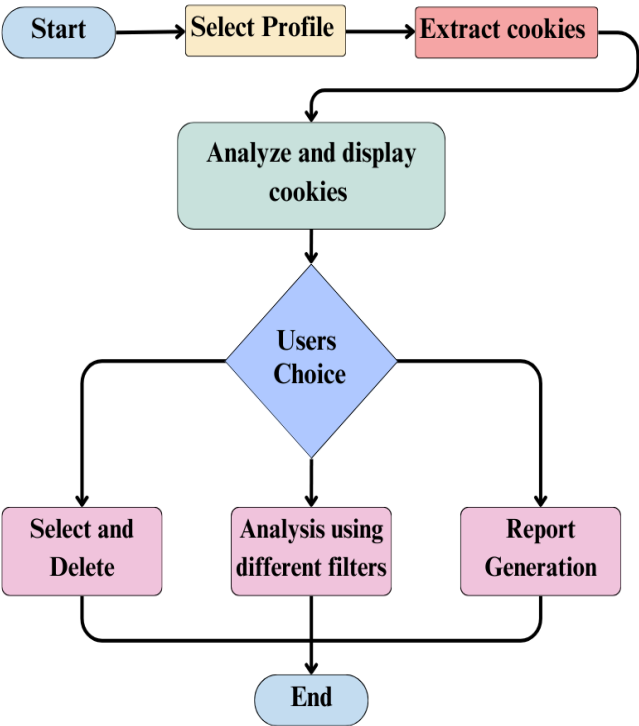


Fig 2. System Architecture

5. Implementation

The implementation of the Persistent Cookie Analyzer with GUI translates the conceptual methodology into a fully functional software application. The Cookie Extraction Module forms the backbone of the tool, utilizing Python’s os and sqlite3 libraries to extract cookies from browser databases. By querying the relevant SQLite tables, attributes such as domain, path, expiration, and security flags are retrieved. The module was tested across multiple browsers to ensure compatibility and accuracy in data extraction.

The Risk Assessment Module operationalizes the scoring framework by analyzing cookies for security vulnerabilities. The scoring algorithm assigns weights to individual risk factors, such as missing Secure/HttpOnly flags (+2 points each) or improper SameSite attributes (+1 point). A cumulative risk score is calculated for each cookie, offering users a quantitative measure of security concerns. The risk assessment results are dynamically updated and displayed in the GUI, providing real-time feedback to users.

The Compliance Checker Module integrates regulatory logic into the tool. Using rule-based algorithms, cookies are evaluated against GDPR, CCPA, and OWASP standards. For example, cookies without user consent are flagged as non-compliant with GDPR, while those with extended expiration periods may fail CCPA compliance. The checker generates recommendations that users can implement to enhance their compliance posture. The Graphical User Interface (GUI) is the interactive layer of the application, designed using Python’s Tkinter framework. It presents cookie data in an organized table format, complete with risk scores and compliance statuses. Interactive buttons allow users to delete or modify cookies, while visual elements such as graphs summarize risk distributions. The interface was designed with usability in mind, ensuring accessibility for non-technical users.

Finally, the Report Generation Module compiles all findings into structured PDF reports using the FPDF library. Reports include cookie details, risk scores, compliance results, and vulnerability summaries, with visual aids to enhance clarity.

This feature is particularly beneficial for organizations seeking documentation for audits or cybersecurity assessments. The implementation was rigorously tested through unit and integration testing to ensure robustness and reliability, with additional packaging into a standalone executable using pyinstaller for ease of deployment.

6. Results

The results of the Persistent Cookie Analyzer highlight its effectiveness in identifying and managing security risks associated with persistent cookies. The tool successfully detects

misconfigured security flags and evaluates vulnerabilities based on expiration times and SameSite attributes. It enables users to sort and filter cookies by risk score, ensuring efficient management. The secure deletion feature allows users to mitigate potential threats without affecting browser functionality. Additionally, the report generation provides detailed insights into cookie vulnerabilities, helping users understand security risks and ensure compliance with privacy regulations like GDPR and CCPA. The tool’s intuitive interface and robust functionality offer a practical solution for managing cookie security and privacy risks.

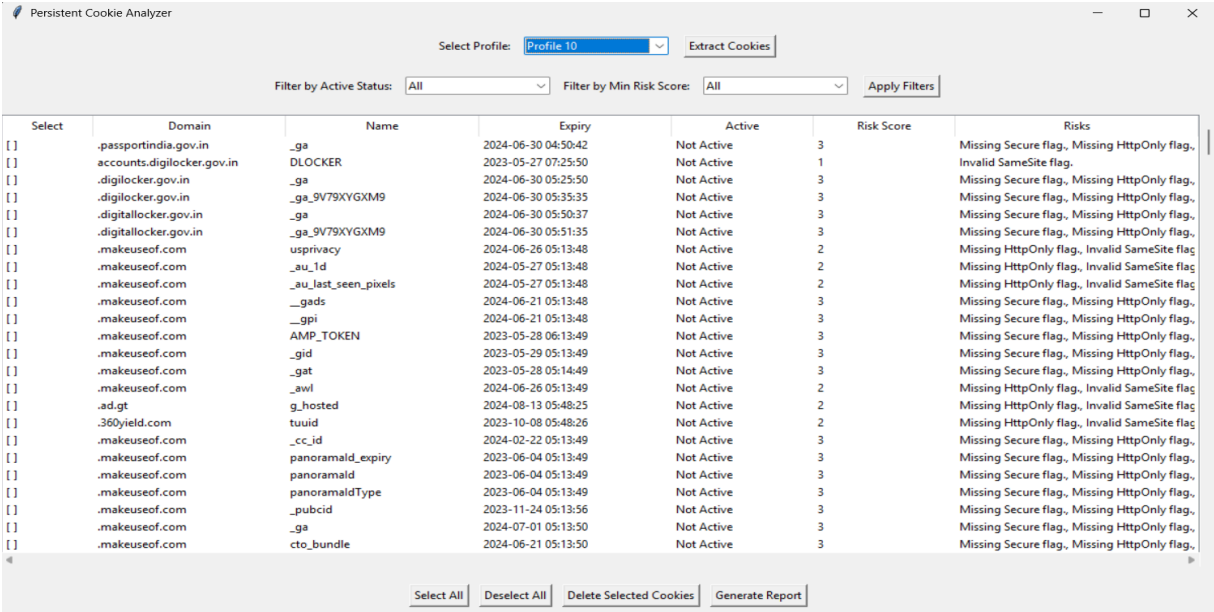


Fig 3. Dashboard

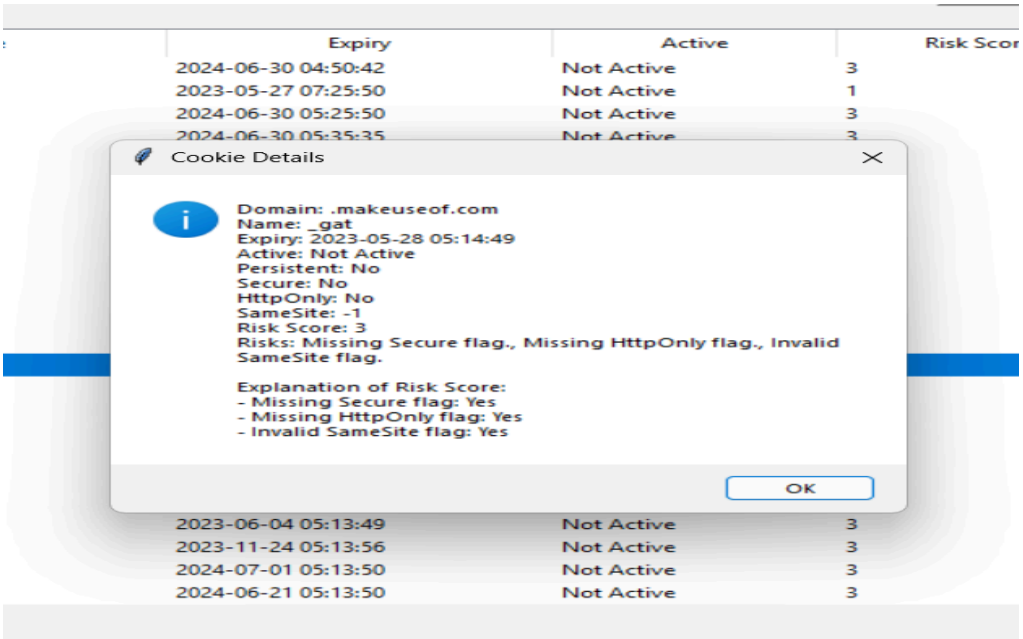


Fig 4. Specifications of the selected cookie

7. Conclusion

In this paper, we presented the design and implementation of the Persistent Cookie Analyzer with a Graphical User Interface (GUI), a tool that facilitates the extraction, analysis, filtering, and deletion of cookies in Google Chrome profiles. The tool addresses critical concerns related to browser privacy and security by assessing cookies for vulnerabilities and enabling users to manage them effectively.

Through the extraction and parsing of cookies from Chrome's SQLite database, followed by a series of security risk assessments, the tool provides a comprehensive approach to safeguarding user data. By incorporating features such as dynamic filtering and a user-friendly GUI, the tool offers an intuitive experience for users to manage and secure their browser cookies.

The performance of the tool has been evaluated through rigorous testing, ensuring that it functions reliably across different user profiles. The flexibility of the tool allows for various filtering criteria, empowering users to tailor the cookie management process according to their needs. Furthermore, the integration of security analysis ensures that cookies which may expose users to risks, such as cross-site scripting (XSS) or man-in-the-middle attacks, are promptly identified and flagged.

In future work, we aim to extend the tool's functionality by adding support for additional browsers, improving the security analysis algorithms, and providing more advanced features for automated cookie management. This tool provides a valuable contribution to enhancing user control over browser privacy, making it an essential tool for individuals concerned about online security.

8. References

- [1] (GDPR), "Regulation (EU) 2016/679 of the European Parliament and of the Council," Official Journal of the European Union, Apr. 2016. [Online]. Available: <https://eur-lex.europa.eu>
- [2] (CCPA), "California Civil Code Title 1.81.5," California Legislative Information, 2018. [Online]. Available: <https://leginfo.legislature.ca.gov>
- [3] OWASP, "OWASP Top Ten Project," Open Web Application Security Project, 2021. [Online]. Available: <https://owasp.org>
- [4] R. Fielding and J. Reschke, "Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content," Internet Engineering Task Force (IETF), RFC 7231, Jun. 2014. [Online]. Available: <https://www.rfc-editor.org>
- [5] B. Richards and L. Wills, "Best Practices for Secure Cookie Management," in Proceedings of the 2020 IEEE International Conference on Cybersecurity (ICCyber), New York, NY, 2020, pp. 243–250.
- [6] A. K. Ghosh, "Web Application Security: A Practitioner's Handbook," IEEE Computer Society, vol. 48, no. 5, pp. 54–63, May 2015.
- [7] Google Developers, "SameSite Cookies," Google Chrome Developer Documentation, Jan. 2020. [Online]. Available: <https://developer.chrome.com>
- [8] S. K. Sharma and P. K. Singh, "A Risk-Based Scoring System for HTTP Cookies," Journal of Cybersecurity Practices, vol. 6, no. 4, pp. 11–22, Oct. 2021.
- [9] Python Software Foundation, "Python 3 Documentation," Python.org, 2024. [Online]. Available: <https://docs.python.org>
- [10] SQLite, "The SQLite Database System," SQLite.org, 2024. [Online]. Available: <https://www.sqlite.org>
- [11] A. Bhattacharya, S. Das, and M. Roy, "Privacy Compliance Assessment for Cookies: A Framework," 2022 IEEE Symposium on Privacy-Aware Computing, New Delhi, India, 2022, pp. 341–348.
- [12] R. Thakur and V. Jain, "Design and Implementation of Browser-Based Cookie Analysis Tools," International Journal of Information Security Research, vol. 10, no. 2, pp. 110–118, Apr. 2022.